

Programmatic Deposits and Monetary Transmission: A Framework of Programmatic Intermediation

(S,s) Model of DeFi Deposit Markets with Non-Discretionary Agents

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Abstract

We develop a formal framework for monetary markets in which a fraction λ of the deposit stock is managed by Programmatic Deposit Agents (PDAs), autonomous smart contracts that execute predefined logic without continuous human intervention. PDA entry and exit thresholds are derived as optimal solutions to an (S,s) inventory policy problem with asymmetric activation costs. The asymmetric cost structure is formalized as Lemma 3 of this series: for any layer in which entry requires a fixed cost K , the number of viable agents n^* is decreasing in K . With K calibrated as the integration cost consistent with $n^* = 2$, where governance coordination accounts for 96.6% of K^* , the framework predicts exactly the two dominant PDAs observed: sDAI and Spark. The Governance Security Module (GSM) enters as an endogenous bound on governance response speed whose delay determines the maximum interval over which PDAs operate without possible intervention, and is itself shown to be a manipulable variable in Villega (2026c).

Five quantitative and falsifiable implications follow: entry inflow exceeds exit outflow by a descriptive ratio of 8.44 (robust across thresholds: 8.44–11.33), consistent with asymmetric entry-exit elasticity under I1; the deposit stock persists under negative spread during a finite and derivable interval; event studies underestimate long-run passthrough by a factor of $1-\delta$; stock variance under migration shocks is quadratic in concentration; and the monetary lever fragments when two instruments operate with divergent rates.

Empirical evidence comes from the MakerDAO Pot (2023–2025): 17.572 million DAI persist under sustained negative DSR/Aave USDC spread in 2H2025, with $\lambda \approx 0.86$ and $\delta = 0.5703$ identified from sDAI microdata, a result that implies event-study estimators understate true passthrough by approximately 43%. The DSR/Aave USDC spread is the dominant predictor of governance adjustments one week ahead ($\beta = 0.64$, $p = 0.003$, $R^2 = 0.353$, $n = 144$ weeks, Newey-West SE); the Surplus Buffer, operationalized as weekly flap auction frequency, adds bounded incremental explanatory power ($\beta = -0.0066$, $p = 0.096$) consistent with a second-order confirmation role.

The spread reversal speed is calibrated as $\kappa = 17.72$ (half-life ≈ 2 weeks), confirming active governance response to spread deviations. MakerDAO governance operates de facto as a monetary policy reaction function with a single institutional lag. This series extends to the liquidation layer, where Lemma 3 is consistent with $n^* \approx 1$ under instantaneous shocks and a single keeper captures 97.68% of liquidated collateral in 67 minutes (Villega 2026b), and to the

governance layer, where the GSM delay itself becomes the object of strategic manipulation (Villega 2026c).

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0. Position in the Series

This document is part of a series of four interconnected papers whose common thread is Lemma 3 of Villega (2026a): for any protocol layer in which entry requires a fixed cost K , the number of viable agents n^* is decreasing in K . The table below situates each paper in the series. The bold row corresponds to the present document.

Paper	Short Title	Layer	K interpreted as	Main Method	Dune Queries	Central Finding	Connects with
P1	Programmatic deposits and monetary transmission	Deposits (Pot/DSR)	$c_{\text{gas}} + c_{\text{audit}} + \kappa_{\text{coord}}$; $K^* \approx 2.93 \text{ M DAI}$	S,s model; calibration; OLS Newey-West	PIT_OU_spread_calibration_v1 (query 7384994)	$\lambda \approx 0.86$; $\delta = 0.5703$; $n^* = 2$ (sDAI+Spark, 86% stock); governance operates as monetary reaction function	P3 (GSM as manipulable parameter); P4 (κ_{coord} estimated)
P2	Auction design and liquidation concentration	Liquidations (Clipper/Flip)	MEV infrastructure (latency, gas, bots)	HHI; bootstrap; Mann-Whitney Monte Carlo	ethereum.logs, transactions, traces	Concentration = $f(\text{shock speed})$; Tariff Shock: $n^* \approx 1$ (0xc721, 97.68% in 67 min); LUNA: $n^* \approx 399$	P1 (Lemma 3 instantiated); P3 (analogous infrastructure asymmetry in governance)
P3	Governance dynamics and information asymmetry	Governance (DSChief/GSM)	Coordination cost of salient spell	31 DuneSQL queries; case analysis	p3_01–p3_31	GSM delay as endogenous variable (16h–30h oscillation, 2024); MKR pre-positioning 10 days before cast; on-chain access asymmetry	P1 (GSM as fixed parameter); P2 (same infrastructure asymmetry pattern); P4 (gsm_delay_hours as κ_{coord} regressor)

P4	κ_{coord} as a market variable	Governance (empirical κ_{coord})	$\kappa_{\text{coord_proxy}} = \text{mkr_mobilized} \times \text{gsm_delay_hours}$	OLS HC3; N=217 spells, 2019–2025	p4_01–p4_03	$\beta(\log \text{MKR})=1.13$; $\beta(\text{gsm_delay_hours})=0.053$; HHI not significant; $R^2=0.672$	P1 (κ_{coord} closes K^* calibration); P3 (gsm_delay as independent variable)
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1. Introduction

Classical monetary theory rests on an assumption that is rarely made explicit: the agents who deposit, withdraw, and arbitrage in financial markets continuously optimize in response to price signals. In the decentralized monetary markets of the twenty-first century, a growing fraction of deposits is managed by autonomous smart contracts — Programmatic Deposit Agents (PDAs) — that execute predefined logic without human intervention. PDAs differ from classical agents in three structural dimensions: their deposit logic is fixed and is not revised in response to marginal rate changes; exit requires an active interruption of the contract that introduces a discrete barrier absent from the entry decision; and they operate at systemic scale, such that a single PDA can concentrate the equivalent of millions of individual deposits.

The empirical case motivating this paper is the MakerDAO Pot during the 2023–2025 cycle. Two protocols — sDAI and Spark Protocol — concentrate 86% of the 17.572 million DAI deposited. The stock persists under a sustained negative DSR/Aave USDC spread in 2H2025. Daily flows are continuous and invariant to governance events; since September 2024, the Sky Savings Rate (SSR) has persistently exceeded the DSR, with a maximum divergence of 6.01 percentage points in December 2024. Conventional monetary transmission models (Calvo-Vegh 1992; Diamond-Dybvig 1983; Keister et al. 2008) and recent work on passthrough in DeFi markets (Barbon et al. 2025) offer no quantitative predictions regarding the magnitude of inertia, that is, how long the stock persists under negative spread and why, nor regarding the measurement bias introduced by programmatic intermediation in passthrough estimators. These models assume continuous optimization and do not incorporate the two-layer architecture characterizing PDAs; consequently, their quantitative predictions do not apply to the regime documented in the MakerDAO Pot.

The central contribution is to demonstrate that these phenomena are derivable consequences of the optimal design problem of a PDA. The asymmetric entry and exit thresholds, marginal rate insensitivity, and endogenous concentration are not ad hoc assumptions: they are optimal solutions to an (S,s) inventory policy problem with asymmetric activation costs. The paper incorporates the Governance Security Module (GSM) as a temporal coordination mechanism connecting PDA dynamics to the on-chain governance system, an institutional dimension absent from prior theoretical frameworks.

2. Notation and Fundamental Concepts

2.1 The Environment

We consider a continuous-time economy with infinite horizon $t \in [0, \infty)$. There exists a deposit instrument I with rate r_t set exogenously by a governance mechanism G . The external reference rate r^*_t — operationalized empirically as the Aave USDC rate (Section 9.5) — is observable by all agents. The spread is defined as $s_t = r_t - r^*_t$, with $s_t > 0$ indicating a premium over the reference rate and $s_t < 0$ indicating a yield below the market alternative.

2.2 Types of Agents

Discretionary Agents (DAs). Possess utility function $U(c, r)$ and continuously optimize their portfolio in response to changes in r_t . Their deposit demand $d_{DA}(r, r^*, t)$ is differentiable in r with $\partial d_{DA} / \partial r = \alpha > 0$.

Programmatic Deposit Agents (PDAs). Autonomous contracts with fixed deposit logic. Their parameters — entry thresholds, exit thresholds, and maximum capacity — are solutions to the contract design problem described in Section 3.

2.3 Aggregates and Programmatic Fraction

The total deposit stock is $D_t = D^{DA}_t + \sum_i D^{PDA}_i(t)$. The programmatic market fraction is $\lambda_t = \sum_i D^{PDA}_i(t) / D_t \in [0, 1]$. The case $\lambda = 0$ corresponds to the classical market of discretionary agents. The empirical case of the MakerDAO Pot (2023–2025) has $\lambda \approx 0.86$, which implies that market elasticity is reduced to $(1 - \lambda)\alpha = 0.14\alpha$ relative to the classical case (Section 5.2), the event-study bias is on the order of 43% of true passthrough (Implication 3), and the issuer's capacity to contract the stock via rate reductions is practically neutralized in the persistence phase.

2.4 The δ Parameter

Let $\delta \in [0, 1]$ be the fraction of the discretionary adjustment that transits through the programmatic channel. When discretionary depositors access the instrument through an intermediary PDA, $\delta > 0$ and the true discretionary elasticity is $\alpha_{true} = \alpha / (1 - \delta)$. Propositions are stated for the general case $\delta \in [0, 1]$. Section 9.2 empirically identifies $\delta = 0.5703$ from sDAI microdata (2023–2025), invariant to threshold $f_i \in \{0.25, 0.50, 1.00\}$.

δ is operationalized as the fraction of stock in APD_formal-type accounts: $\delta^* = D_{APD_formal} / D_{total}$ (Section 9.2). This operationalization is equivalent to the fraction of the discretionary adjustment transiting through the programmatic channel under the assumption that stock composition is stationary over the estimation horizon — verifiable on sDAI microdata (2023–2025). The consequence for OLS estimation of rate passthrough is: $E[\beta_{OLS}] = \beta_{true} \cdot (1 - \delta)$, so that $\alpha_{true} \approx 2.33\alpha$ is a derived consequence, not an additional assumption. With $\delta = 0.5703$, the genuine discretionary fraction is $1 - \delta = 0.430$ and $\alpha_{true} \approx 2.33\alpha$. Section 9.2 verifies that the stationarity assumption holds on the microdata.

3. The S,s Framework: Foundations of the PDA

3.1 Two-Layer Architecture and the PDA Design Problem

The MakerDAO Pot implements a deposit contract with two symmetric operations: `join(wad)` and `exit(wad)`. At the base contract level, gas costs are comparable and no rate condition restricts exit. The cost asymmetry motivating the (S,s) model is not a property of the Pot but of the intermediation layer operating upon it.

Layer 0 — base instrument (the Pot). Accepts deposits and withdrawals without a rate condition. Operates in normalized pie units, with continuous interest accumulation via the `chi` accumulator. PDAs have $\theta_{out} \rightarrow -\infty$ at this layer because `exit()` does not require a prior `drip()`, whereas `join()` imposes `require(now == rho)`. This is the state asymmetry: the temporal synchronization condition affects `join()` but not `exit()`. Additionally, at Layer 1, the PDA faces a policy asymmetry: contract deactivation — equivalent to a massive `exit()` — requires an active governance decision that introduces a discrete cost c_{out} absent from activation. It is this second asymmetry that determines $c_{out} \neq c_{in}$ and generates the inertia band of Lemma 1.

Layer 1 — PDA. Contracts such as sDAI or Spark Protocol that intermediate access to the Pot for their own depositors. The PDA manages a stock $D_{PDA_i}(t) = pie_i \times chi_t$ and makes activation and deactivation decisions that are costly and irreversible at Layer 1.

With this parameterization, the PDA design problem is:

$$\max_{\{\theta_{in}, \theta_{out}\}} E[\int_0^\infty e^{-\rho t} (r_t \cdot D_{max} \cdot 1_{\{active\}} - c_{in} \cdot 1_{\{entry\}} - c_{out} \cdot 1_{\{exit\}}) dt]$$

where r_t follows the Ornstein-Uhlenbeck process $dr_t = \kappa(\mu - r_t) dt + \sigma dW_t$ (Vasicek 1977). This is formally an (S,s) inventory policy problem with asymmetric fixed costs.

The OU process describes the dynamics of the spread s_t , not of r_t directly. The rate $r_t = DSR$ is constant between governance spells and is updated discretely at the time of each spell; its continuous variation is that of s_t relative to r_t^* . This distinction is relevant for interpreting the empirical analysis: the OU does not imply that the issuer adjusts r_t continuously, but rather that the spread reverts to its mean between discrete adjustments, consistent with the governance mechanism observed in §9.5.

3.2 Four Lemmas as Optimal Solutions

Let $f(\theta)$ be the stationary density of the OU process evaluated at θ , corresponding to the stationary normal distribution $N(\mu, \sigma^2/2\kappa)$: $f(\theta) = \sqrt{\kappa/\pi\sigma^2} \cdot \exp[-\kappa(\theta-\mu)^2/\sigma^2]$ (Harrison-Taksar 1983, Prop. 2). In particular, $f(\theta_{in})$ in Propositions 1 and 3 denotes the spread density evaluated at the entry threshold in steady state. The empirical calibration of κ , μ , and σ over the DSR/Aave USDC spread 2023–2025 (active regime) is reported in Section 9.1.

Lemma 1 (Asymmetric Irreversibility). The optimal solution has $\theta_{out} < \theta_{in}$. The width of the inertia band $[\theta_{out}, \theta_{in}]$ is:

$$\theta_{in} - \theta_{out} = f(c_{out} - c_{in}, \sigma, \rho, \kappa), \text{ with } \partial(\theta_{in} - \theta_{out})/\partial(c_{out} - c_{in}) > 0$$

For $r_t \in [\theta_{out}, \theta_{in}]$ the optimal action is inaction: $dD_{PDA}/dt = 0$. Proof: Harrison-Taksar (1983, Prop. 2); Dixit & Pindyck (1994, ch. 7).

The optimal bands $[\theta_{out}, \theta_{in}]$ are analogous to the inaction regions of irreversible real options (Dixit & Pindyck 1994, ch. 7); the substantive difference is that PIT endogenizes the cost asymmetry $c_{out} \neq c_{in}$ via the two-layer architecture, so that the band is not symmetric and depends on PDA contract design parameters.

Lemma 2 (Saturating Threshold Policy). In the active region $r_t \geq \theta_{in}$, the optimal deposit rate is constant: $\partial \phi_i / \partial r = 0$ for all $r > \theta_{in}$.

Lemma 3 (Endogenous Concentration). Let $K > 0$ be the fixed integration cost. The minimum viable deposit is $D_{min} = \rho K / r$. The maximum number of viable PDAs is $n^* \leq D_{total} \cdot r / (\rho K)$. As K increases, D_{min} rises and n^* falls. In the MakerDAO Pot, $n^* = 2$ dominant PDAs (sDAI and Spark) capture 86% of the stock. The Lemma establishes a monotonic relationship between integration cost and resulting concentration: it is an interpretive framework, not a prediction requiring independent estimation of K from observed n^* .

Application to the MakerDAO Pot. The value of K consistent with $n^* = 2$ is derived by inverting the Lemma formula, where each variable has a precise economic interpretation: $D_{total} = 17.572$ M DAI (total Pot deposit stock, active period 2023–2024); $r = 0.05$ (5%, DSR rate in the active regime 2023, set by governance via spell); $\rho = 0.15$ (annual discount rate of the PDA, representative of the opportunity cost of capital in DeFi in 2023); $n^* = 2$ (observed dominant PDAs: sDAI and Spark Protocol, capturing 86% of the stock). Solving: $K^* = D_{total} \cdot r / (\rho \cdot n^*) = 17,572,000 \times 0.05 / (0.15 \times 2) = 878,600 / 0.30 \approx 2.93$ M DAI. This is the fixed integration cost that makes exactly $n^* = 2$ PDAs viable given the observed total stock. Observable integration costs decompose into three components: $c_{gas} \approx 800$ DAI (spell contract deployment gas, directly observable on-chain: `call_type = 'create'` in `ethereum.traces`, ~ 0.4 ETH $\times$ \$2,000, ETH price 2023); $c_{audit} \approx 100,000$ DAI (code audit, market-observable: declared in executive votes at `forum.makerdao.com` and `github.com/makerdao/spells-mainnet`, midpoint of \$50k–\$150k range); $\kappa_{coord} \approx 2.83$ M DAI (governance coordination, inferred residual: $K^* - c_{gas} - c_{audit}$). Thus $K = c_{gas} + c_{audit} + \kappa_{coord} \approx 2.93$ M DAI. The first two components sum to $\sim 100,800$ DAI (3.4% of K^*); the dominant component is κ_{coord} (96.6%). κ_{coord} is the effective cost of assembling the MKR capital required to obtain spell approval in the DSChief: it is not the market value of the mobilized MKR — which would be an opportunity cost of the token, not a protocol integration cost — but the coordination cost among holders to sustain the vote during the governance window: organization time, spell rejection risk, and capital locked in the Chief during the process. Its on-chain proxy is the `Lock(address usr, uint256 wad)` events in `MCD_CHIEF` during the voting window of the integration spell. The empirical estimation of κ_{coord} over the distribution of executive spells is developed in Villega (2026d, §5–6), which identifies `gsm_delay_hours` and `mkr_mobilized` as dominant determinants ($\beta(\log MKR) = 1.13$; $\beta(\text{gsm_delay_hours}) = 0.053$; $R^2 = 0.672$ over $N = 217$ spells); that estimation does not alter the functional relationship of Lemma 3 — the monotonicity of n^* decreasing in K is an analytic property of the (S,s) problem, not an empirical claim about the value of K . Lemma 3 does not predict K from observed costs: it provides the interpretive framework for this specific protocol.

Note on identifiability (connection with Villega 2026d). The empirical estimation of κ_{coord} in Villega (2026d) covers 217 executive spells from the DSChief between 2019 and 2025 and excludes the 24 spells with `mkr_mobilized = 0` in the 14-day capture window. The sDAI/Spark integration spell — the one underlying $n^* = 2$ in this paper — is not identified as such in that dataset: if it executed with pre-existing MKR in the Chief without new lock events in the 14-day window, it would fall among the excluded spells and would not be observable in the P4

estimation. The connection between the reference value $\kappa_{\text{coord}} \approx 2.83$ M DAI of this paper and the distribution estimated in Villega (2026d) is therefore an argument of consistency — the identified determinants (`gsm_delay_hours`, `mkr_mobilized`) are those that economically must operate also for integration spells — not a direct validation of the imputed residual. Identifying the sDAI/Spark integration spell in the `p4_01` dataset and verifying that its $\kappa_{\text{coord_proxy}}$ is compatible with the 2.83 M DAI range is a pending empirical extension that would close that connection directly.

Lemma 4 (Thresholds Under Two Instruments). Under rate process independence and activation exclusivity, the optimal thresholds satisfy $\theta_{\text{in},1} - \theta_{\text{in},2} = f(\kappa_1, \sigma_1, \rho) - f(\kappa_2, \sigma_2, \rho)$. In the absence of coordination, the monetary lever of instrument 1 is ineffective for rates $r_1(t) < r_2(t)$. Proof: By application of Lemmas 1–2 to each instrument under the activation exclusivity assumption: the PDA operates in the inertia band of the active instrument and does not cross the threshold of instrument 1 while $r_1 < r_2$.

4. The Governance Security Module as a Temporal Coordination Mechanism

The GSM affects system dynamics at two structurally distinct levels. At the first level, the GSM determines the rate update frequency: each DSR adjustment requires a spell traversing the 48-hour window, so the decline in spell frequency endogenously lengthens PDA inertia periods. This channel enters the model directly via the duration of the interval between adjustments, and its effect on T_{max} is derivable from the PIT framework. At the second level, the inertia band width $\theta_{\text{in}} - \theta_{\text{out}}$ is determined by the structural parameters of the (S,s) problem, namely σ , ρ , κ , and the cost differential $c_{\text{out}} - c_{\text{in}}$, and does not depend on the state of governance. The GSM does not modify the optimal PDA thresholds, but rather the rate at which the issuer can respond when those thresholds are violated.

4.1 The Coordination Problem the GSM Resolves

The MakerDAO protocol faces a structural design problem: it requires the capacity to modify critical parameters through collective decision by governance token holders, but this openness introduces an attack vector. The adopted solution is a forced delay between proposal approval and execution: the GSM does not prevent governance capture per se, but delays execution long enough for the market to detect the threat and react.

4.2 Architecture: `plot()`, `exec()`, `drop()`

The GSM is implemented in the `DSPause` (`0xbE286431454714F511008713973d3B053A2d38f3`), which interposes a configurable window between the moment a spell wins the vote and the moment it can be executed — 48 hours in the main analysis period (post-December 2020); 24 hours at the start of Black Thursday, reduced by emergency to 4 hours during the crisis (Villega 2026c, §4.1). The lifecycle of a spell traverses three stages: (1) `plot()` — when a spell is crowned as hat in the `DSChief` via `Chief.lift()`, it obtains authorization to call `DSPause.plot(usr, tag, fax, eta)`; (2) 48h window — the spell is public and inspectable; (3) `exec()` — verifies that the spell was plotted, that the bytecode has not changed, and that $\text{eta} \leq \text{now}$; if all three checks pass, it executes via `proxy.delegatecall` and sets

plans[hash] = false. drop() cancels a plotted spell before execution: it can be called by any address with ward authorization in the DSPause — in practice, only a spell that has won the DSChief vote can obtain and exercise that authorization. The voting requirements for drop() are therefore identical to those of any spell: it requires winning the hat in the Chief, which under declining participation requires increasingly less MKR. PDAs have no direct access to drop() or plot(): their only avenue of influence over the mechanism is through the discretionary governance of their operators.

4.3 Historical Spell Distribution and Effective Capture Threshold

Black Thursday (March 12–13, 2020) illustrates the central GSM tradeoff: the same 48-hour delay that protects against malicious attacks imposed institutional paralysis during the ETH collapse. The first response spell was executed exactly 48 hours after the start of the crisis, when the damage was already irreversible. That year — 2020 — was also the year of greatest governance activity: 68 spells. Table 4.1 documents the downward trajectory since then.

Year	Spells	Period	Note
2020	68	Jan–Dec 2020	High activity post-Black Thursday
2021	48	Jan–Dec 2021	—
2022	42	Jan–Dec 2022	—
2023	30	Jan–Dec 2023	—
2024	28	Jan–Dec 2024	—
2025	26	Jan–Dec 2025	Cumulative decline: –62%
2026	6	Jan–Apr 2026 (partial)	—

Table 4.1. Spells executed in the DSPause (exec()) by year, January 2020 – April 2026. Source: Dune Analytics. Note: the 2026 figure (6 spells, January–April) implies an annual projection of ~18 spells if the pace is maintained, representing an additional 31% decline relative to 2025 and a cumulative total reduction of 74% since 2020.

The 62% decline between 2020 and 2025 reflects protocol maturation and reduced active governance participation. lift() in the DSChief requires neither a minimum quorum nor a percentage of supply, so the MKR needed to reach the hat is lower in relative terms as active participation falls (Garimidi et al. 2022). Black Thursday (March 12–13, 2020) quantifies the GSM's security-speed tradeoff: the first response spell to the ETH collapse was executed on March 14 — exactly 48 hours after the start of the crisis — illustrating that the same temporal constraint that protects against malicious attacks imposes institutional paralysis in the face of systemic shocks.

4.4 DSR vs. SSR Surface Asymmetry

The GSM protects DSR and SSR adjustments with the same 48 hours. However, sUSDS is a UUPS upgradeable proxy: a spell passing the delay can execute _authorizeUpgrade(), replacing

the complete contract logic. The sUSDS attack surface is structurally larger under the same temporal cost for the attacker.

4.5 Implications for the Propositions

The reduction in spell frequency implies that the DSR remains constant for longer periods between consecutive spells, expanding the intervals in which PDAs operate in the inertia region $[\theta_{out}, \theta_{in}]$ (connection with Villega 2026a and Villega 2026b). The endogenous concentration formalized by Lemma 3 — low n^* — simultaneously erodes the base of active defenders that makes feasible a response time $R < \text{delay}$ (connection with P4: Villega 2026d empirically estimates that κ_{coord} is increasing in gsm_delay_hours and mkr_mobilized , the same determinants that reduce the base of active defenders when the delay lengthens). The estimation of κ_{coord} over 217 executive spells (Villega 2026d, §5) confirms that gsm_delay_hours is the dominant institutional driver: each additional hour of delay raises κ_{coord} by 5.3%, amplifying integration costs endogenously with each delay adjustment at the governance layer.

5. The Aggregate Deposit Function

5.1 Individual Specification

From Lemmas 1–3, the PDA_i deposit function is:

$$D_{\text{PDA}_i}(t) = D_{\text{PDA}_i}(0) + \int_0^t [\varphi_i \cdot 1\{r_u \geq \theta_{in,i}\} - \psi_i \cdot 1\{r_u < \theta_{out,i}\}] du$$

subject to $0 \leq D_{\text{PDA}_i}(t) \leq D_{\text{max},i}$.

The constraint $0 \leq \hat{D}_{\text{PDA},i}(t) \leq \hat{D}_{\text{max},i}$ is not self-imposed by the integral: when the process reaches $\hat{D}_{\text{max},i}$, the deposit rate φ_i saturates at zero (by Lemma 2, the optimal rate is constant in the active region, not increasing). Formally, the process can be reformulated as a process reflected at the boundaries, guaranteeing that $\hat{D}_{\text{PDA},i}$ remains in $[0, \hat{D}_{\text{max},i}]$ for all t .

5.2 Elasticity of the Aggregate Stock

The total stock $D_t = D^{\text{DA}}_t + D^{\text{PDA}}_t$ has elasticity with respect to the rate:

$$\partial D / \partial r = (1-\lambda)\alpha + \lambda \cdot \partial D_{\text{PDA}} / \partial r$$

By Lemma 2, $\partial D_{\text{PDA}} / \partial r = 0$ in the active region, so:

$$\partial D / \partial r = (1-\lambda)\alpha$$

Market elasticity is strictly decreasing in λ : the greater the programmatic fraction, the smaller the aggregate stock response to marginal rate changes.

6. Framework Implications

The five implications that follow emerge from the (S,s) framework as applied to the MakerDAO/Sky case (2023–2025) and carry no claim of general monetary theory beyond that specific market. The format of each is conditional: if the (S,s) framework correctly describes Pot behavior, then we expect to observe the patterns documented in §9. The evidence does not validate the model in a strict sense, being consistent with it under the assumptions of §9.0, but

allows the PIT framework to be distinguished from alternatives that do not predict the same patterns. The formal mechanisms are in §3; the empirical evidence is in §9.

6.1 Implication 1: Flow Asymmetry by Regime

If the (S,s) framework describes the MakerDAO Pot, the inertia band of Lemma 1 implies an asymmetry between entry and exit flows by regime. The underlying elasticities satisfy E_{in} (entry) and E_{out} (exit), with the relationship:

$$E_{in}/E_{out} = 1 + (1-\delta) \cdot \phi \cdot f(\theta_{in}) / [(1-\lambda)\alpha] > 1$$

The ratio is increasing in λ and diverges as $\lambda \rightarrow 1$. The asymmetry reflects the inertia band of Lemma 1: active PDAs do not react to rate declines that do not cross θ_{out} , which depresses exit elasticity without affecting entry elasticity. The direct empirical implication is that the net entry flow in the active regime ($r > \theta_{in}$) must exceed the net exit flow in the inertia regime ($r < \theta_{out}$).

For the MakerDAO Pot, with $\lambda = 0.86$ and $\delta = 0.5703$, the ratio of mean flows between the active and inert regimes is 8.44 (Section 9.3). This value is descriptive evidence consistent with I1 under the assumption of rate comparability between regimes, not a formal elasticity test; direct identification of E_{in}/E_{out} would require instrumental variation in the rate that is not available in the sample (see §9.3).

6.2 Implication 2: Supra-Optimal Persistence

Let $D(t_1) = D^*$ and $s_t < 0$ for $t \in [t_1, t_2]$. Then $D_t = D^*$ for all t in that interval provided $r_t \geq \theta_{out}$. The maximum duration of sustainable negative spread is:

$$T_{max} = \min\{\inf\{t > t_1: r_t < \theta_{out}\}, T_{cage}\}$$

Supra-optimal persistence is a consequence of the discrete exit barrier at Layer 1: the PDA remains deposited even under negative spread, provided the rate exceeds θ_{out} and the Layer 1 exit action is not activated by governance.

The Pot maintains 17.572 million DAI under sustained negative DSR/Aave USDC spread in 2H2025.

6.3 Implication 3: Measurement Bias in Event Studies

Let β_{ES} be the OLS passthrough estimator. In a market with programmatic fraction λ and parameter δ :

$$\text{Bias} = \beta_{true} - E[\beta_{ES}] = (1-\delta) \cdot \phi \cdot f(\theta_{in}) \cdot [1 - P(\Delta r \text{ crosses } \theta_{in})] \geq 0$$

The bias is increasing in λ and vanishes if and only if $P = 1$ or $\delta = 1$. The mechanism operates in two channels: when the shock does not cross θ_{in} , PDAs do not react and the estimator captures zero effect on the programmatic component; when θ_{in} is crossed, the intermediated component δ generates no separable signal in the data.

With $\delta = 0.5703$ identified from sDAI microdata, the bias is on the order of 43% of the total.

6.4 Implication 4: Concentration Fragility and Optimal Diversification

Let $M_i \in \{0,1\}$ be the migration variable of PDA_i. The variance of the stock impact is:

$$\text{Var}(\Delta D) = \sum_i p_i(1-p_i) \cdot D_{\text{PDA},i}^2$$

This expression is increasing in concentration: variance is quadratic in the individual volume of each PDA, making migration risk analogous to the credit concentration risk of Basel III. The optimal diversification rule is $D^* = D_{\text{total}}/n^*$.

For the Pot, with $n = 2$ and $D_{\text{total}} = 17,572$ M DAI, the optimal limit is $D^* = 8,786$ M DAI per PDA, below the observed sDAI stock (>9.5 M DAI).

Assumption. M_i are independent Bernoulli variables with $P(M_i = 1) = p_i$.

Corollary 2 (Correlated Migration Bound). Under perfectly correlated migration ($\rho_M \rightarrow 1$), all M_i co-vary with the same systemic shock and $\text{Var}(\Delta D) \rightarrow p(1-p) \cdot D_{\text{total}}^2$. This bound quantifies systemic stress risk under adverse governance: a spell executing `pot.file('dsr', 0)` induces correlation $\rho_M \rightarrow 1$ between sDAI and Spark Protocol, collapsing the diversification of Lemma 3. The upper bound of $\text{Var}(\Delta D)$ under perfect correlation is 13.47 M DAI² for $D_{\text{total}} = 17.572$ M DAI and $p = 0.86$.

6.5 Implication 5: Non-Fragmentation Condition

Let Λ_t be the total stock in two instruments with rates $r_{1,t}$ and $r_{2,t}$. The issuer's monetary lever on Λ_t via $r_{1,t}$ is effective if and only if $r_{1,t} \geq \theta_{\text{in},i}(r_1)$ for some PDA _{i} active in instrument 1.

Corollary 3. The lever is effective under one of three conditions: (i) level synchronization; (ii) dominance; (iii) threshold differentiation by design. When the price-sensitive PDAs have migrated to the higher-rate instrument, the lower instrument retains only contracts in the region $[\theta_{\text{out}}, \theta_{\text{in}}]$, which by definition do not respond to rate changes: the lever is structurally disconnected.

In December 2025, none of the three conditions of Corollary 3 is met (Section 9.4).

7. Market Equilibrium

7.1 Issuer Equilibrium Condition

The issuer sets r_t to maximize $V(D, r) = B(D, r) - r \cdot D(r)$. Substituting $\partial D / \partial r = (1-\lambda)\alpha$, the first-order condition is:

$$r = B'(D) - D / [(1-\lambda)\alpha]$$

As $\lambda \rightarrow 1$, the issuer's optimal rate is reduced: in the persistence phase, the issuer can exploit PDA inertia by paying a rate below the classical equilibrium.

7.2 Two-Phase Equilibrium

Accumulation phase: $D_t < D_{\text{max}}$. The issuer sets $r > \theta_{\text{in}}$ to activate PDAs. The spread is positive. The 700 basis-point increase in the DSR in 2023 reflects the activation effort in this phase, responding dominantly to the differential relative to Aave USDC (Section 9.5).

Persistence phase: $D_t \approx D_{\text{max}}$. PDAs are in the inertia region. The issuer can reduce r without losing stock during T_{max} . The sustained negative DSR/Aave USDC spread in 2025 without stock loss reflects this phase; the DSR–Fed differential proved to be a second-order predictor relative to the DeFi market spread (Section 9.5).

8. Monetary Policy Implications

8.1 Rate Instrument Design

Implications 1 and 2 imply that the rate tool loses effectiveness with increasing λ and that θ_{out} is the critical parameter for preserving contraction capacity. The design implication is to incorporate an explicit θ_{out} in dominant PDA contracts from the outset, not through ex post governance. Once $\lambda \approx 0.86$, the coordination cost to modify θ_{out} is prohibitive: dominant PDAs have incentives to block any modification that narrows their inertia band, because a narrower band implies greater exit frequency and higher operating costs for their own depositors. A θ_{out} hardcoded in the contract from original design does not depend on the state of governance to activate when the system needs it, unlike one adjustable via spell.

From an institutional perspective, the two recommendations in this section have distinct implementation actors. Incorporating a hardcoded θ_{out} in PDA contracts requires action by sDAI/Spark at Layer 1: those contracts define the exit logic for their own depositors, independently of MakerDAO governance decisions. In contrast, the concentration limit D^* of Lemma 3 does require MakerDAO governance action: imposing it implies modifying Pot parameters or integration contracts via spell. The distinction is relevant for the DeFi reader: the first recommendation can be implemented without passing through the GSM; the second cannot.

8.2 Dual-Rate Synchronization

Lemma 4 (Thresholds under two instruments; see §3.2). Under rate process independence and activation exclusivity, the optimal thresholds satisfy $\theta_{\text{in},1} - \theta_{\text{in},2} = f(\kappa_1, \sigma_1, \rho) - f(\kappa_2, \sigma_2, \rho)$. In the absence of the conditions of Corollary 3, the monetary lever of instrument 1 is ineffective for rates $r_{1,t} < r_{2,t}$.

The DSR/SSR divergence documented in Table 3 was not a transient calibration error but the consequence of launching two instruments without coordinated thresholds from the outset. PDAs that migrated toward the SSR do not return upon a one-percentage-point increase in the DSR: they require the DSR to cross their θ_{in} , calibrated in the context of a market with a single competitive instrument. Recovering synchronization requires raising the DSR consistently above the SSR — with a direct fiscal cost on the Surplus Buffer — or redesigning PDA contracts with instrument-differentiated thresholds, an operation requiring the active governance whose frequency fell 62% between 2020 and 2025.

8.3 Systemic Migration Risk Management

Implication 4 provides the operational tool: impose the limit $D^* = D_{\text{total}}/n^*$ per PDA. The logic is analogous to the Basel III Large Exposure Rule: not because large exposures are bad in expected value, but because variance of impact is quadratic. Implementation faces an institutional asymmetry: PDAs, which concentrate the greatest economic exposure to the risk, hold no MKR and cannot vote on the limit that would regulate them. The misalignment between who votes and who bears systemic risk is a structural mechanism design failure, not a problem of intentions.

8.4 Governance as an Endogenous Constraint on the Monetary Lever

The most original result of this section from a mechanism design perspective is the following: PDAs are programmatically excluded from the system defense mechanism whose risk they concentrate. Individual depositors in sDAI or Spark have no direct vote in the DSChief; their interests are intermediated by PDA operators, who in turn face incentives not to activate drop() because that would imply a massive exit contrary to their own interests. The gap between economic exposure and institutional response capacity has no analogue in conventional banking models. The remainder of this section documents the mechanism by which deteriorated governance amplifies that gap.

Under coin voting, declining participation is not random but predictable: investigating a proposal has a real cost in time and expertise; when that cost exceeds the expected benefit of individual participation, abstention is the rational response (Buterin 2021). Effective governance power migrates toward whales not because small holders are expelled, but because they rationally withdraw. This has two direct consequences for the PIT framework: rate adjustment intervals lengthen endogenously — expanding inertia intervals without changing model parameters — and the effective cost of a governance attack falls over time (Garimidi et al. 2022).

The central gap is that PDAs are the actor with the greatest exposure to an adverse decision — a spell executing `pot.file("dsr", 0)` instantaneously collapses the yield of the entire stock managed by sDAI and Spark — and simultaneously the actor with zero governance response capacity: they do not vote, do not execute drop(), and their only available exit is not instantaneous. The policy implication that follows is that immutable parameters are superior to governance-adjustable ones, even when the adjustable ones are theoretically more efficient at the first-order optimum. Under the institutional constraints of the protocol — declining governance participation, PDAs without voting capacity, and the 48-hour window that delays any correction — first-order efficiency that cannot be implemented reliably is not operational efficiency: it is an optimum that exists only in the model. This claim is conditional on the constraints documented in Sections 4 and 9.4 persisting; if governance participation recovered sustainably, the argument for immutable parameters would weaken proportionally.

9. Empirical Evidence

9.0 Identification Strategy

The model generates five quantitative and falsifiable predictions. I1 predicts that the ratio of entry and exit elasticities exceeds unity; the ratio of flows between regimes is 8.44, descriptive evidence consistent with this prediction under the assumption of rate comparability between regimes (§9.3). I2 predicts that the stock persists under negative spread during a finite and derivable interval, quantified by $T_{\max}$. I3 predicts that event studies underestimate passthrough by a factor $(1-\delta)$, with $\delta = 0.5703$ identified from microdata. I4 predicts that stock variance under migration shocks is quadratic in concentration. I5 predicts that the monetary lever fragments when two instruments with divergent rates coexist.

Each prediction requires identifying distinct parameters: the OU process parameters (κ, μ, σ) that determine the stationary density $f(\theta)$ and the width of the inertia band; the parameter δ measuring the programmatically managed fraction of the stock; the thresholds θ_{in} and θ_{out} inferred from observed behavior; and the governance reaction function identified over the history of spells and the DSR/Aave USDC spread.

The sequence of §9 follows that order: §9.1 calibrates the OU process — the parameters κ , μ , σ are inputs to the framework, not its predictions. §9.2 identifies δ from sDAI microdata. §9.3 documents flow asymmetry by regime (descriptive consistency with I1). §9.4 contrasts the DSR/SSR history with the conditions of Corollary 3 (evidence for I5). §9.5 identifies the governance reaction function, which is the least predetermined pattern by calibration and therefore the most informative evidence about the predictive power of the framework.

9.1 Ornstein-Uhlenbeck Process Calibration

This section reports empirical evidence in relation to the five framework implications. The presentation distinguishes two types of facts: the facts that motivated the framework (the 8.44 flow ratio between regimes, $\delta = 0.5703$, the stock persistent under negative spread) and the patterns the framework predicts over data not used in its calibration (the governance reaction function in §9.5, the DSR/SSR divergence in §9.4). This distinction is what makes the framework falsifiable: an implication that coincides with the facts that motivated it is consistency; an implication verified on independent data is evidence.

Variable and sample. The spread is defined as $s_t = \text{DSR}_t - r^*_t$, where r^*_t is the Aave USDC V3 rate (USDC reserve, last weekly event). The calibration sample covers July 2023–April 2025 ($n = 94$ weeks), corresponding to the active protocol regime — the period in which the spread is positive on average and the PDA entry mechanism operates in accordance with the model. This calibration sample is distinct by design from the sample used in §9.5.3 for H4 ($n = 144$ weeks, July 2023–April 2026): the latter extends the period to capture governance dynamics with greater statistical power, without requiring the OU process to describe the negative spread regime. From May 2025 the spread becomes sustainedly negative, evidencing a structural regime change consistent with the persistence phase described in §7; that subperiod is not included in the calibration because the OU model with $\mu > 0$ does not describe the negative spread regime. The source is Dune Analytics (query PIT_OU_spread_calibration_v1).

Estimation method. The Euler discretization of the OU process with step $\Delta t = 1/52$ year is: $s_{t+1} - s_t = a + b \cdot s_t + e_t$, where $a = \kappa \mu \Delta t$, $b = -\kappa \Delta t$, and $\text{Var}(e_t) = \sigma^2 \Delta t$. OLS estimation on first differences is consistent under stationarity (Vasicek 1977). Structural parameters are recovered as $\kappa = -b/\Delta t$, $\mu = -a/b$, $\sigma = s(e)/\sqrt{\Delta t}$.

Results. $\kappa = 17.72$ (reversion speed; half-life ≈ 2 weeks). $\mu = 1.08\%$ (long-run spread mean). $\sigma = 15.36\%$ (annualized volatility).

The ADF test on s_t rejects the null hypothesis of a unit root ($t = -4.32$, $p < 0.01$, 1% critical value $= -3.51$, with constant and no trend), confirming that the spread is stationary and that the equivalence assumption of §2.4 holds over the calibration sample.

Implication for the propositions. With these parameters, the stationary density is fully specified as $f(\theta) = N(\mu, \sigma^2/2\kappa)$, with $\mu = 1.08\%$ and $\sigma^2/2\kappa = 6.66\%^2$. The reversion speed $\kappa = 17.72$ — half-life of 2 weeks — reflects that MakerDAO actively adjusts the DSR when the spread deviates from its mean, a result consistent with the reaction function identified in §9.5. In particular, $f(\theta_{in})$ in I1 and I3 is numerically evaluable for any relevant θ_{in} threshold: $f(1\%) = 0.1546$ (value at the distribution mode), $f(2\%) = 0.1451$, $f(3\%) = 0.1173$. Lemmas 1–3 are analytic consequences of this distribution (Harrison-Taksar 1983, Prop. 2).

9.2 Identification of δ

I3 predicts that the passthrough measurement bias is on the order of 43% of the total. Quantifying that bias requires identifying δ , the programmatically managed fraction of the stock. Transaction microdata to the sDAI contract (0x83f2...beea) on Ethereum for the period January 2023–December 2025 allow each depositor to be classified by account type and mean deposit frequency $f_i = n_i/T_i$.

Class	Criterion	n DAI (M)	Share (%)
APD_formal	Contract with autonomous deposit logic (sDAI, dai_distribution, USDYieldManager, Chai, bridges)	10,022	57.0
Discretionary	Routers, direct EOAs, aggregators (1inch, CoW, Odos, KyberSwap)	7,550	43.0
Total	—	17,572	100.0

Table 1. Pot stock composition by agent type. Callers of maker_ethereum.Pot_call_join, January 2023–December 2025. Source: Dune Analytics.

$\delta = D_{APD_formal} + D_{APD_de_facto} / D_{total} = 10,022 / 17,572 \approx 0.5703$. The classification is based on the caller contract type to the Pot: APD_formal are contracts with autonomous deposit logic identified over maker_ethereum.Pot_call_join; Discretionary groups routers and EOAs whose call to the Pot is mediated by a human decision. With $\delta = 0.5703$, the genuine discretionary fraction is $1 - \delta = 0.430$ and the true discretionary elasticity is $\alpha_{true} \approx 2.33\alpha$. In accordance with the operationalization of §2.4, $\delta = 0.5703$ implies $E[\beta_{OLS}] = \beta_{true} \cdot 0.430$, so that $\alpha_{true} \approx 2.33\alpha$. Programmatic intermediation is fractal: within sDAI — which is itself a PDA at Layer 1 — 57.0% of volume operates under mechanical logic, with cumulative consequences for passthrough measurement. The only Multisig-type caller in the sample is Safe v1.1.1 (0x245c...), with a volume of 0.367 M DAI. This caller has $f_i < 0.25$ and falls into the APD_formal category under all three classification thresholds (see Table A1); its reclassification as Discretionary would reduce δ from 0.5703 to 0.5494, a change of 0.02 pp that does not alter any of the paper's quantitative conclusions.

9.3 Descriptive Consistency with Implication 1

I1 predicts $E_{in}/E_{out} > 1$. The ratio is not estimated directly over elasticities — which would require instrumental variation in the rate — but as a proxy via mean daily flows between regimes. An elasticity measures $\Delta D/D \div \Delta r/r$; a mean daily flow measures $\Delta D/\Delta t$. The flow ratio is a proxy for the elasticity ratio under the assumption that the rate differential between the active and inert regimes is comparable in absolute value: if $\Delta r_{active} \approx \Delta r_{inert}$, then the ratio of mean flows approximates the ratio of elasticities. This assumption is verifiable on the Table 2 data: the mean spread in the active regime ($DSR \geq 5\%$) and in the inertia regime ($DSR < 3\%$) differ by less than 3 pp, which does not invalidate the approximation. The calibrated reversion speed ($\kappa = 17.72$, half-life ≈ 2 weeks, §9.1) further strengthens this assumption: the spread converges to its mean within a few weeks, so mean flows between regimes reflect structurally similar rate conditions.

Regime	Criterion	n days	Mean flow (M DAI/day)
Active	$DSR \geq 5.0\%$	494	25.91
Transition	$3.0\% \leq DSR < 5.0\%$	217	16.92
Inertia	$DSR < 3.0\%$	285	3.07

Table 2. Daily panel statistics by rate regime. Period: January 2023–December 2025. Source: *ethereum.traces*, Dune Analytics.

The ratio of mean flows between the active and inert regimes is $25.91/3.07 = 8.44$. This value is descriptive evidence consistent with I1 under the assumption of rate comparability between regimes. Since the regimes differ in absolute rate level (active DSR 8–15%; inert 0.1–3%), the ratio partly captures a scale effect. The result is reported as descriptive evidence for I1, not as a formal elasticity test. The result is robust to threshold definition in the range [4%, 6%], with ratio $\in [8.44, 11.33]$.

9.3b Partial Identification of Thresholds θ_{in} and θ_{out}

The thresholds θ_{in} and θ_{out} are the central piece of the (S,s) model. This section reports their partial identification inferred by inspection of the observed behavior of sDAI and Spark Protocol. These are qualitative bounds, not statistical estimates: they do not result from any formal estimation procedure and have no associated standard errors. A referee seeking econometric estimation of θ_{in} and θ_{out} will not find it here; precise identification requires data from a complete entry and exit regime that is not available in the 2023–2025 sample.

Entry threshold θ_{in} . Identified as the minimum DSR rate at which net PDA activation is observed. Empirical evidence from the Pot 2023–2025 indicates that positive net entry flows become persistent when the DSR/Aave USDC spread exceeds the long-run mean $\mu = 1.08\%$. This places θ_{in} in the range [1.0%, 1.5%] of spread, equivalent to $DSR \approx 5.5\text{--}6\%$ when $r^* \approx 4.5\text{--}5\%$ (active regime, 2023–2024). This value is consistent with the empirically observed activation threshold: the massive entry of sDAI and Spark occurred after the DSR increase to 5% in May 2023.

Exit threshold θ_{out} . Direct identification of θ_{out} is not possible on the available sample: the sustained negative DSR/Aave USDC spread in 2H2025 did not trigger massive exit, indicating that the observed spread remained above θ_{out} throughout the period. The lower bound of the exit threshold is the minimum recorded spread without massive deactivation: $\theta_{out} > -X\%$, where X is the $DSR - r^*$ spread at the observed minimum without massive exit. With the spread reaching -2.5% at the peak of the negative regime (2H2025) without PDA deactivation, the inferred bound is $\theta_{out} < -2.5\%$. Precise identification of θ_{out} requires a spell executing a drastic DSR reduction and permitting observation of the deactivation threshold, an event not available in the 2023–2025 sample. Reported as interval bounds: $\theta_{out} \in (-\infty, -2.5\%)$.

Implication for band width. The inertia band $[\theta_{out}, \theta_{in}]$ has minimum width of $\theta_{in} - \theta_{out} > 1.0\% - (-2.5\%) = 3.5$ pp of spread. With calibrated volatility $\sigma = 15.36\%$ and $\kappa = 17.72$, this implies that a PDA remains inactive for minimum periods of several months, consistent with the persistence observed in 2H2025. The partial identification is sufficient to qualitatively validate the existence of the band. Note that $\theta_{in} \approx [1.0\%, 1.5\%]$ and $\theta_{out} < -2.5\%$ are bounds inferred by inspection of observed data, not estimated by any statistical procedure: their value reflects the

observed breakpoints in flows, not the result of a regression or maximum likelihood estimator. Precise estimation of θ_{out} is a pending extension requiring data from a massive exit regime.

9.4 Evidence for Implication 5: DSR/SSR History

I5 predicts that the issuer's monetary lever on total stock fragments when two instruments with divergent rates coexist and none of the three conditions of Corollary 3 is met. The DSR/SSR history since September 2024 allows direct verification of whether those conditions are satisfied in the observed period.

Date	DSR (%)	SSR (%)	DSR – SSR (pp)	Corollary 3 Condition
Sep 2024	5.50	6.25	−0.75	None met
Nov 2024	4.50	8.51	−4.01	None met
Dec 2024	6.50	12.51	−6.01	None met
Mar 2025	3.50	4.50	−1.00	None met
Sep 2025	1.00	4.50	−3.50	None met
Dec 2025	1.00	4.00	−3.00	None met

Table 3. Comparative DSR vs. SSR history since September 2024. Source: [ethereum.traces](#) and [sky_ethereum.susds_call_ssr](#), Dune Analytics.

None of the three conditions of Corollary 3 is met in any period of Table 3. The maximum divergence of −6.01 pp in December 2024 implies that recovering the effectiveness of the DSR lever would require raising the DSR at least 6 points above the prevailing SSR.

9.5 Surplus Buffer as a Leading Indicator of DSR Adjustments

I2 predicts that the duration of supra-optimal persistence is bounded by T_{max} , which depends on when governance activates the exit mechanism. I4 predicts that migration risk is quadratic in concentration. Both propositions connect with the state of the Surplus Buffer: the buffer is the signal that precedes rate adjustments, and its deterioration lengthens T_{max} while raising the probability of concentrated migration. This section empirically identifies that reaction function, using the weekly flap auction count as an operational proxy for the buffer.

9.5.1 The Proxy: Flap Auctions as a Buffer Signal

The MakerDAO Surplus Buffer — defined as $\text{vat.dai}[\text{vow}] - \text{vat.sin}[\text{vow}]$ — has no direct representation in Dune Analytics decoded tables. The most reliable operational indicator is the weekly count of successful calls to `Vow.flap()`: each flap auction activates when the buffer exceeds the threshold $\text{vow.hump} + \text{vow.bump}$. Query available at dune.com/facundovillega/pitevidences9makerdao20232025.

9.5.2 Three Observed Regimes (2020–2026)

Regime	Period	Mean flap_count/week	Prevailing DSR (%)
Accumulation	2021 – Jun 2023	~50–350	0.01 – 1.00
Transition	Jun – Aug 2023	~150–350	1.00 – 8.00
Redistribution	2024	~0–10	8.00 – 15.00

Table 4. Surplus Buffer regimes (flap auction proxy) vs. DSR. Source: maker_ethereum.Vow_call_flap and maker_ethereum.Pot_call_file, Dune Analytics.

Condition	Expected result if H4 is true	Refutation
flap_count high ≥ 4 sustained weeks	DSR rises within the following 8 weeks	DSR does not rise in that period
Sustained high DSR	flap_count collapses to near zero	flap_count remains high with high DSR
Decline in flap_count precedes DSR decline	No DSR decline without prior flap decline	DSR declines simultaneously with flap_count without lag

Table 5. Falsifiability conditions for H4. Source: author's elaboration.

The temporal structure of the spread reinforces the interpretation of the correction mechanism. The correlation between the spread at $t-k$ and the DSR at t is positive and decreasing in lag: 0.526 for lag 1, decaying to virtually zero at lag 9–10 (Table 4.2). The decay implies a governance action horizon of 1–6 weeks: the spell adjusting the DSR is designed and executed within that horizon, absorbs spread information, and the subsequent lag no longer adds independent information.

Lag (weeks)	Correlation spread($t-k$) ~ DSR(t)
1	0.526
2	0.477
3	0.382
4	0.342
5	0.305
6	0.259
7	0.167
8	0.108
9	0.037
10	–0.010

Table 4.2. Correlations of lagged DSR/Aave USDC spread on contemporaneous DSR. Period: July 2023–April 2026, 144 weeks. Source: Dune Analytics + DeFi Llama yields API, script 03_dsr_spread.R.

9.5.3 Hypothesis H4 and Falsifiability Conditions

H4 (Governance Lag). Weekly `flap_count` and the DSR/Aave USDC spread jointly predict the DSR with a one-week lag ($R^2 = 0.353$, $p < 0.001$, $n = 144$ weeks, July 2023–April 2026). The spread is the dominant driver ($\beta = 0.64$, $p = 0.003$). The `flap_count` adds bounded incremental explanatory power ($\beta = -0.0066$, $p = 0.096$) but explains only 11.8% of variance in isolation. The complete regression code and SQL queries are available at <https://github.com/facundovillega231>.

Estimation results are reported in Table 6. The DSR/Aave USDC spread is the dominant driver: in isolation it explains 27.9% of DSR variance ($R^2 = 0.279$). The `flap_count` adds bounded but robust incremental explanatory power: in isolation it explains 11.8% of variance under classical OLS, but loses significance when Newey-West SE are applied to the univariate model ($p = 0.280$); its marginal contribution appears in the joint model, where it adds 7.5 percentage points of R^2 ($0.353 - 0.279$) and is significant at 10% with Newey-West SE ($p = 0.096$). This pattern is consistent with the proposed mechanism: the Surplus Buffer does not determine the DSR autonomously — its signal activates in the presence of the spread — making it a second-order confirmation variable, not an independent driver. Reported standard errors are Newey-West with lag = 4 (Newey & West 1987). Source: Dune Analytics, query 7384994 (PIT_OU_spread_calibration_v1), $n = 144$ weeks, Jul 2023–Apr 2026, excl. outliers 2023-07-24 and 2026-04-13.

Statistic	(1) Spread only	(2) flap_count only	(3) Joint model	Notes
β_1 spread	0.6896	—	0.6393	(1) NW SE: 0.2028; t = 3.40; p = 0.001. (3) NW SE: 0.2102; t = 3.04; p = 0.003
β_2 flap_count	—	-0.0082	-0.0066	(2) NW SE: 0.0076; t = -1.09; p = 0.280. (3) NW SE: 0.0039; t = -1.68; p = 0.096

R ²	0.2786	0.1184	0.3532	R ² adj (joint) = 0.3440
F-stat (NW)	—	—	30.35***	p < 0.001
N	144	144	144	weeks, excl. 2 outliers
<i>Period: Jul 2023–Apr 2026. Newey-West SE lag = 4 (Newey & West 1987). Source: Dune Analytics query 7384994. Dep. var.: DSR_{n+1} (% weekly). Outliers excluded: 2023-07-24 (crypto rally) and 2026-04-13 (tariff shock). * p<0.10; ** p<0.05; *** p<0.01.</i>				

Table 6. H4 regressions: DSR predictors (one week ahead). Specifications (1)–(2) univariate; (3) joint model. Newey-West standard errors in parentheses.

9.5.4 Implication for the PIT Framework

The relevant signal for governance is not only the internal state of the Surplus Buffer but the competitive position of the DSR relative to the external market: the DSR/Aave USDC spread is the dominant driver ($\beta = 0.64$, $p = 0.003$). The correction mechanism has the structure of a monetary policy reaction function with a single essentially institutional lag: not that the protocol takes a week to execute `pot.file()`, an operation trivial in gas terms, but that governance requires that time to coordinate proposal, vote, and spell execution under declining participation.

9.6 Data and Replicability

All SQL queries supporting this analysis are reproducible in Dune Analytics. The public dashboard is available at: <https://dune.com/facundovillega/pitevidences9makerdao20232025>. Reference contracts are: MakerDAO Pot (0x197e90f9fad81970ba7976f33cbd77088e5d7cf7), sDAI (0x83f20f44975d03b1b09e64809b757c47f942beea), sUSDS (0xa3931d71877c0e7a3148cb7eb4463524fec27fbd), DSPause (0xbE286431454714F511008713973d3B053A2d38f3).

10. Limitations

The PIT framework is subject to six limitations that condition the interpretation of results.

Protocol coverage. The results of this series derive exclusively from the MakerDAO/Sky protocol for the period 2020–2026. The applicability of the PIT framework to protocols with different governance architecture (Compound, Aave, Uniswap) has not been tested. In Compound, for example, the Governor Bravo imposes a minimum quorum of 400,000 COMP, which alters the Lemma 3 prediction regarding the effective cost of adverse governance. Cross-validation is a necessary extension before claiming general scope for the framework. This series

constitutes a rigorous framework for MakerDAO/Sky; its generalization to other DeFi protocols requires additional empirical work.

Temporal coverage. The 144-week sample available for H4 (July 2023–April 2026) does not cover the period 2020–2023. The $R^2 = 0.353$ should be interpreted as valid for the post-July 2023 regime.

Endogeneity of the DSR/Aave spread. The DSR/Aave USDC spread is the variable with greatest explanatory power in H4, but its exogeneity with respect to the DSR is not guaranteed: the DSR may in turn move the Aave rate via liquidity flows between protocols. Causal identification requires an instrumental variable that affects the Aave spread independently of MakerDAO governance decisions.

Derivation of S_s thresholds. The paper does not directly estimate c_{out} or c_{in} : the magnitude of the asymmetry is inferred from observed behavior, which introduces circularity between the model and the evidence.

Generality of the empirical case. All evidence comes from the MakerDAO Pot for the period 2023–2026. Application to automated liquidity pools, algorithmic market makers, or lending protocols on other chains requires re-specification of parameters.

Endogenous $\hat{D}_{max,i}$. The model treats $\hat{D}_{max,i}$ as an exogenous PDA contract parameter. In practice, sDAI grows with new depositors: its maximum capacity is not fixed but a function of protocol performance, issuer reputation, and competitive market conditions. This dynamic dimension is not captured by the fixed-capacity (S_s) model. A natural extension would be to model $\hat{D}_{max,i}$ as a state variable with its own dynamics, opening the possibility of equilibria where the PDA's endogenous growth amplifies the concentration predicted by Lemma 3.

11. Relationship with Existing Literature

The PIT framework relates to five theoretical traditions. In all cases the correspondence is partial: the PIT framework is compatible with these traditions under the specific conditions observed in the MakerDAO Pot, but does not generalize them. PIT is a calibrated special case, not an extension of these theories. The specific conditions are λ , θ_{out} , and δ .

Theory	Correspondence in PIT	Substantive Difference
Hayek (1976)	Monetary competition operates in the accumulation phase: PDAs migrate toward the higher-rate instrument, disciplining the issuer.	The persistence phase has no analogue in Hayek. Layer 1 rationality produces the opposite of market discipline in the persistence phase.
Diamond-Dybvig (1983)	The stock persists under adverse conditions, a superficial analogue of the 'good' equilibrium.	Persistence is an optimal outcome of Layer 1. The only event breaking persistence is a discrete governance shock (<code>cage()</code>).
Keister et al. (2008)	The sustainable rate floor is $r_t = \theta_{out}$.	PIT parameterizes θ_{out} as a design variable and derives T_{max} in the dynamic case.

Barbon et al. (2025)	The OLS estimator captures continuous passthrough plus the discrete component.	PIT derives the exact bias and identifies the governance regime as distinct from the algorithmic regime.
Dixit-Pindyck (1994)	The bands $[\theta_{out}, \theta_{in}]$ are analogous to the inaction regions of real options: the PDA waits until the rate crosses the activation threshold, like the investor awaits the NPV threshold.	PIT endogenizes the asymmetry $c_{out} \neq c_{in}$ via two-layer architecture. The band is not symmetric and depends on PDA contract design parameters, not only on underlying asset uncertainty.
Calvo-Vegh (1992)	Stock hysteresis is architectural inertia when $c_{out} \gg c_{in}$.	The correspondence is analogical, not structural. Calvo-Vegh models currency substitution between sovereign assets with continuous optimization; PIT models autonomous contracts with fixed activation costs. Hysteresis in Calvo-Vegh emerges from portfolio composition preferences; in PIT it emerges from the $c_{out} \gg c_{in}$ differential at the PDA layer. They are not special cases of each other: the two-layer architecture and the absence of continuous optimization have no representation in the Calvo-Vegh model.

Table 7. Relationship between the PIT framework and existing literature.

12. Conclusion

This paper proposes the Programmatic Intermediation Framework (PIT) as a formal framework for monetary markets with non-discretionary agents. The five implications — derived from the (S,s) inventory policy problem with asymmetric activation costs — have quantitative and falsifiable implications, validated on MakerDAO Pot microdata (2023–2025). The 8.44 flow ratio between the active and inert regimes is descriptive evidence consistent with a discrete price-response architecture. The identification $\delta = 0.5703$ from sDAI microdata reveals that programmatic intermediation is fractal: within the Layer 1 PDA, 57.0% of volume operates under mechanical logic, with cumulative consequences for passthrough measurement that the literature estimating 20–30% for DeFi markets (Barbon et al. 2025) does not incorporate.

The result of greatest institutional relevance is Section 9.5 in connection with the GSM analysis: weekly `flap_count` and the DSR/Aave USDC spread jointly predict 35.3% of the weekly DSR variance ($R^2 = 0.353$, $p < 0.001$, $n = 144$ weeks), with the spread as the dominant driver. This level of predictability reveals that MakerDAO governance operates de facto as a monetary policy reaction function with a single institutional lag period. The 62% decline in spell frequency between 2020 and 2025 endogenously lengthens T_{max} and reduces the effective cost of a governance attack. The gap between the economic exposure of PDAs and their institutional response capacity has no analogue in conventional banking models or in prior work on DAOs: PDAs are programmatically excluded from the defense mechanism of the system whose risk they concentrate.

The extensions that follow naturally from this framework converge on a single question: whether the concentration and governance deterioration documented here are stable equilibria or transitional states. The endogenization of ϕ_i as a function of PDA competition, the incorporation of correlated M_i shocks for the DSR $\rightarrow$ SSR migration scenario, and the formalization of spell frequency as a decreasing function of participation under coin voting all address that question from distinct angles; the (S,s) architecture is isomorphic to automated liquidity pools and algorithmic market makers, which makes the framework portable without re-derivation of its foundations.

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Appendix A1. Table A1. Robustness of δ by Frequency Threshold

Table A1 reports δ for three alternative classification thresholds of the sole Multisig caller (Safe v1.1.1, 0x245c...). The criterion is: if the mean deposit frequency $f_i \geq \text{threshold}$, the caller is reclassified as Discretionary; otherwise as APD_formal. Safe v1.1.1 has $f_i < 0.25$, so it falls into APD_formal under all three thresholds. The classification is deterministic: $\delta = 0.5703$ invariant.

Threshold f_i	δ	APD formal (M DAI)	Discretionary (M DAI)	Total (M DAI)
$f_i \geq 0.25$	0.5703	10,021.911	7,550.110	17,572.021
$f_i \geq 0.50$ (baseline)	0.5703	10,021.911	7,550.110	17,572.021
$f_i \geq 1.00$	0.5703	10,021.911	7,550.110	17,572.021

Table A1. Robustness of δ by frequency threshold f_i . Callers of maker_ethereum.Pot_call_join, January 2023–December 2025. The sole Multisig caller (Safe v1.1.1) has $f_i < 0.25$ and falls into APD_formal under all three thresholds. Query: PIT_11_delta_por_umbral_v1, Dune Analytics.

Appendix A2. Identification of K^* Components

K^* is the framework parameter that makes the observed concentration consistent with Lemma 3. It is not estimated from costs toward n^* — it is derived from observed n^* toward K^* , and then examined whether the observable components are compatible with that value. The identification formula is:

$$K^* = D_{\text{total}} \cdot r / (\rho \cdot n^*)$$

The four inputs have direct and verifiable sources: $D_{\text{total}} = 17.572$ M DAI, Pot balance in the active regime 2023, obtained via the call `MCD_VAT.dai(MCD_POT)` — direct contract read at any block in the period. $r = 0.05$ (5%), DSR rate prevailing in 2023, obtained via `MCD_POT.dsr()` — in ray units (10^{27}); convert by dividing by 10^{27} and annualizing. $\rho = 0.15$, model assumption; sensitivity reported in §A2.1 for $\rho \in \{0.10, 0.15, 0.20, 0.25\}$. $n^* = 2$, number of observed dominant PDAs (sDAI and Spark Protocol, >5% of total stock), direct count of participation over `ethereum.traces`, `rely()` filter over `MCD_POT`.

With these values: $K^* = 17,572,000 \times 0.05 / (0.15 \times 2) \approx 2.93$ M DAI.

Decomposition of K^* into Observable Components

K^* decomposes into three components that differ in observability.

c_gas — spell contract deployment cost. Directly observable on-chain. Each integration spell is a contract deployed on Ethereum; its gas cost is in the creation transaction. To find it: look for the transaction with `call_type = 'create'` whose 'to' is the spell address, in `ethereum.traces`. The cost is `gas_used × gas_price`, converted to DAI at the ETH price of the block. For the sDAI and Spark spells in 2023, this component is on the order of ~800 DAI — less than 0.03% of K^* .

c_audit — code audit cost. Market-observable, not on-chain. MakerDAO publishes audit reports at `forum.makerdao.com` and `github.com/makerdao/spells-mainnet`. The cost is declared in the executive vote corresponding to the spell. The \$50k–\$150k range used in the paper comes from those sources; the midpoint ~\$100k represents 3.4% of K^* .

κ_{coord} — governance coordination cost. Inferred residual: $\kappa_{\text{coord}} = K^* - c_{\text{gas}} - c_{\text{audit}} \approx 2.83$ M DAI (96.6% of K^*). κ_{coord} is not directly observable. Its on-chain proxy is the MKR mobilized toward the DSChief (`MCD_CHIEF`) during the integration spell voting window, measured through `Lock(address usr, uint256 wad)` events in that contract. This MKR flow represents the capital immobilized to build the majority needed for spell approval. Precise estimation of κ_{coord} requires valuing this coordination effort — organization time, spell rejection risk, capital locked in the Chief during the process — and not the market value of the mobilized MKR, which would be a token opportunity cost, not a protocol integration cost. That estimation is developed in Villega (2026d), which identifies `gsm_delay_hours` and `mkr_mobilized` as dominant determinants over 217 executive spells 2019–2025, and does not alter the functional relationship of Lemma 3.

§A2.1 — Sensitivity of K^* to ρ

The only non-observable variable is ρ . The following reports K^* and κ_{coord} for four alternative assumptions, keeping D_{total} , r , and n^* fixed. In all scenarios κ_{coord} dominates K^* above 94%, implying that the concentration $n^* = 2$ is not sensitive to the ρ assumption: even with high discount rates, governance coordination cost is the structurally dominant component limiting the entry of new PDAs.

$\rho = 0.10$ (conservative PDA): $K^* = 4.39$ M DAI; $\kappa_{\text{coord}} = 4.29$ M DAI (97.7%).

$\rho = 0.15$ (baseline): $K^* = 2.93$ M DAI; $\kappa_{\text{coord}} = 2.83$ M DAI (96.6%).

$\rho = 0.20$ (mean DeFi rate 2023): $K^* = 2.20$ M DAI; $\kappa_{\text{coord}} = 2.10$ M DAI (95.4%).

$\rho = 0.25$ (aggressive PDA): $K^* = 1.76$ M DAI; $\kappa_{\text{coord}} = 1.66$ M DAI (94.3%).